Fourier Series: Projection, Summation, and Regularity

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The Fourier series of a periodic function is simultaneously a set of coordinates, a way of reconstructing a signal, and a spectral decomposition of differentiation. These interpretations agree, but only if we distinguish three notions of convergence: convergence in mean square, pointwise convergence, and uniform convergence. A single function with a jump already forces the distinction.[1]

§1.1 Fourier coefficients are orthogonal coordinates

Work on the circle $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ with the normalized inner product

$$\langle f, g\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\overline{g(x)}\,dx.$$

For $n \in \mathbb{Z}$, put $e_n(x) = e^{inx}$. Then

$$\langle e_m, e_n\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{i(m-n)x}\,dx = \delta_{mn}.$$

Thus the characters e_n form an orthonormal system; for $f \in L^2(\mathbb{T})$ define

$$\boxed{\widehat{f}(n) = \langle f, e_n\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}\,dx.}$$

The truncated Fourier series is

$$S_N f(x) = \sum_{|n|\le N} \widehat{f}(n)e^{inx}.$$

It is the orthogonal projection of f onto the $(2N+1)$-dimensional space of trigonometric polynomials of degree at most N. Indeed, for every such polynomial P,

$$\langle f - S_N f, P\rangle = 0,$$

and Pythagoras gives the best-approximation identity

$$\boxed{\|f - P\|_2^2 = \|f - S_N f\|_2^2 + \|S_N f - P\|_2^2.}$$

With this normalized inner product, the norm and Fourier coefficients will obey Parseval's formula without any extra 2π factors.

[1] The normalization, Fejér argument and applications follow Stein and Shakarchi, *Fourier Analysis: An Introduction*, Chapters 1–3.

§1.2 Fourier Coefficients versus Point Values

Take $f(x) = x$ on $(-\pi, \pi)$, extended 2π-periodically. It is odd, so its cosine coefficients vanish. Integration by parts gives, for $n \geq 1$,

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^\pi x \sin(nx)\, dx \\ &= \frac{2}{\pi} \left[-\frac{x\cos(nx)}{n} + \frac{\sin(nx)}{n^2} \right]_0^\pi = \frac{2(-1)^{n+1}}{n}. \end{aligned}$$

Consequently,

$$\boxed{x \sim 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nx).}$$

For $-\pi < x < \pi$, the series converges to x. At $x = \pi$, the periodic extension jumps from π to $-\pi$; the series takes the midpoint 0, since every term is zero there.

The coefficients decay only like $1/n$. This slow decay is a trace of the jump. Compare the continuous even function $g(x) = |x|$. Again by integration by parts,

$$a_0 = \pi, \qquad a_n = \frac{2}{\pi} \int_0^\pi x \cos(nx)\, dx = \frac{2((-1)^n - 1)}{\pi n^2}.$$

Thus

$$\boxed{|x| = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\cos((2k+1)x)}{(2k+1)^2}.}$$

The corner has produced coefficients of order $1/n^2$, and the series now converges uniformly by the Weierstrass test. Evaluating at $x = 0$ yields

$$\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}.$$

Since the odd terms form three quarters of $\sum_{n\geq 1} n^{-2}$, this also gives the Basel sum $\sum_{n\geq 1} n^{-2} = \pi^2/6$.

§1.3 The Dirichlet kernel explains both power and danger

The partial sum has a convolution representation:

$$S_N f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t) D_N(t)\, dt,$$

where

$$\boxed{D_N(t) = \sum_{n=-N}^{N} e^{int} = \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)}.}$$

The integral of D_N is 2π, but it is not nonnegative, and its L^1 norm grows on the order of $\log N$. Ordinary partial sums are therefore not averages with positive weights. Oscillation can help reconstruct a function at a point while simultaneously allowing overshoots near a jump.

For a piecewise C^1 periodic function, Dirichlet's convergence theorem gives

$$\lim_{N\to\infty} S_N f(x) = \frac{f(x+0) + f(x-0)}{2}.$$

This theorem is consistent with the sawtooth calculation; it does not assert uniform convergence in the presence of jumps.

In fact, the overshoot has an exact limiting profile. For the sawtooth, put $x_N = \pi - t/N$ with fixed $t > 0$. The sine identity $\sin(n(\pi - y)) = (-1)^{n+1}\sin(ny)$ yields

$$S_N f(x_N) = 2\sum_{n=1}^{N} \frac{\sin(nt/N)}{n} = \frac{2}{N}\sum_{n=1}^{N} \frac{\sin(tn/N)}{n/N}.$$

This is a Riemann sum, so

$$\boxed{\lim_{N\to\infty} S_N f(\pi - t/N) = 2\int_0^t \frac{\sin u}{u}\, du.}$$

At $t = \pi$, the limit is approximately 3.7039, exceeding the left-hand value π by about 0.5623. Relative to the jump height 2π, the overshoot is about 8.95%. Its spatial width shrinks like $1/N$, but its height does not vanish: this is the Gibbs phenomenon.

§1.4 Fejér averaging repairs the kernel

Instead of taking the last partial sum, average all the partial sums through N:

$$\sigma_N f = \frac{1}{N+1}\sum_{k=0}^{N} S_k f.$$

Its convolution kernel is

$$\boxed{F_N(t) = \frac{1}{N+1}\left(\frac{\sin((N+1)t/2)}{\sin(t/2)}\right)^2.}$$

Unlike D_N, the Fejér kernel satisfies

$$F_N(t) \geq 0, \qquad \frac{1}{2\pi}\int_{-\pi}^{\pi} F_N(t)\, dt = 1.$$

Moreover, for any fixed $\delta > 0$,

$$\sup_{\delta \leq |t| \leq \pi} F_N(t) \leq \frac{1}{(N+1)\sin^2(\delta/2)}.$$

The mass of the kernel concentrates near $t = 0$.

For continuous periodic f, let $\omega_f(\delta) = \sup_{|x-y|\leq\delta} |f(x) - f(y)|$, with distance understood on the circle. Positivity allows us to estimate

$$\begin{aligned} |\sigma_N f(x) - f(x)| &\leq \frac{1}{2\pi}\int_{-\pi}^{\pi} F_N(t)|f(x-t) - f(x)|\, dt \\ &\leq \omega_f(\delta) + \frac{2\|f\|_\infty}{(N+1)\sin^2(\delta/2)}. \end{aligned}$$

First choose δ so the modulus of continuity is small, then choose N large. The bound is uniform in x, proving

$$\boxed{\|\sigma_N f - f\|_\infty \longrightarrow 0.}$$

Fejér's theorem is therefore more than a different summation convention: it changes a sign-changing convolution kernel into a positive approximate identity.

§1.5 Parseval and the precise meaning of completeness

Trigonometric polynomials are uniformly dense in the continuous periodic functions, by Fejér's theorem. Continuous functions are dense in $L^2(\mathbb{T})$, so trigonometric polynomials are dense in $L^2(\mathbb{T})$. Together with orthogonality, this proves that e_n, $n \in \mathbb{Z}$, form an orthonormal basis.

Consequently the orthogonal projections satisfy $S_N f \to f$ in L^2 for every $f \in L^2(\mathbb{T})$, and Parseval's identity is

$$\boxed{\frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x)|^2\,dx = \sum_{n\in\mathbb{Z}} |\widehat{f}(n)|^2.}$$

For the sawtooth, its complex coefficients satisfy

$$|\widehat{f}(n)| = \frac{1}{|n|} \quad (n \neq 0).$$

Therefore

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} x^2\,dx = \frac{\pi^2}{3} = 2\sum_{n=1}^{\infty}\frac{1}{n^2}.$$

This is a second, independent route to the Basel identity, now through orthogonal projection rather than evaluating a uniformly convergent cosine series.

Notice what Parseval does *not* claim: the Fourier series need not reproduce the prescribed point value of f at a discontinuity. An L^2 function is an equivalence class modulo changes on null sets; its Hilbert-space coordinates are insensitive to isolated point values.

§1.6 Regularity and the Sobolev threshold

For a sufficiently regular periodic function, integration by parts gives

$$\widehat{f'}(n) = in\widehat{f}(n).$$

Each derivative multiplies the n-th Fourier coefficient by in. This motivates the Sobolev norm

$$\boxed{\|f\|_{H^s(\mathbb{T})}^2 = \sum_{n\in\mathbb{Z}} (1+n^2)^s |\widehat{f}(n)|^2.}$$

It is initially defined for trigonometric polynomials and extends to distributions for which the sum is finite.

For $s > 1/2$, Cauchy–Schwarz gives

$$\begin{aligned}\sum_{n\in\mathbb{Z}} |\widehat{f}(n)| &\le \left(\sum_{n}(1+n^2)^{-s}\right)^{1/2}\left(\sum_{n}(1+n^2)^{s}|\widehat{f}(n)|^2\right)^{1/2}\\ &\le C_s\|f\|_{H^s}.\end{aligned}$$

The first sum is finite exactly when $s > 1/2$. Hence every $H^s(\mathbb{T})$ function with $s > 1/2$ has an absolutely and uniformly convergent Fourier series, determining a continuous representative. This proves the one-dimensional Sobolev embedding

$$H^s(\mathbb{T}) \hookrightarrow C^0(\mathbb{T}), \qquad s > \frac{1}{2}.$$

The threshold is not an arbitrary label: it comes directly from the summability of inverse frequency weights.

§1.7 Heat and waves act differently on the same coordinates

Fourier characters diagonalize the Laplacian:

$$-\partial_x^2 e^{inx} = n^2 e^{inx}.$$

For initial data $f \in L^2(\mathbb{T})$, the periodic heat equation $u_t = \kappa u_{xx}$ has solution

$$\boxed{u(x,t) = \sum_{n\in\mathbb{Z}} e^{-\kappa n^2 t}\,\widehat{f}(n)e^{inx}}, \qquad t > 0.$$

For every positive time, the Gaussian decay in n beats every polynomial growth factor n^m. The heat solution is therefore smooth in x, even if its initial data have jumps.

For the wave equation $u_{tt} = c^2 u_{xx}$ with initial displacement f and velocity g, each nonzero Fourier mode solves a harmonic oscillator equation. The resulting formula is

$$\begin{aligned} u(x,t) = {} & (\widehat{f}(0) + t\widehat{g}(0)) \\ & + \sum_{n\neq 0} \left(\widehat{f}(n)\cos(c|n|t) + \frac{\widehat{g}(n)}{c|n|}\sin(c|n|t) \right) e^{inx}. \end{aligned}$$

The heat semigroup damps high frequencies; wave propagation rotates their phases without smoothing their amplitudes. The same orthogonal basis detects a fundamental difference between irreversible diffusion and reversible oscillation.

§1.8 Fourier series can represent distributions

Point masses are not L^2 functions, but their Fourier coefficients are meaningful by duality. For the periodic Dirac distribution at zero, define

$$\langle \delta_0, \varphi \rangle = \varphi(0)$$

on smooth periodic test functions. Its Fourier coefficients in the integral normalization above are all $1/(2\pi)$, so formally

$$\boxed{\delta_0 = \frac{1}{2\pi}\sum_{n\in\mathbb{Z}} e^{inx}.}$$

This equality is true as an equality of periodic distributions: pairing either side with a smooth test function gives $\varphi(0)$. It is not a pointwise sum of ordinary functions.

The distributional viewpoint explains why frequency methods continue to work for singular initial data and fundamental solutions. Orthogonal projection controls mean-square approximation, positive kernels control uniform summation, Sobolev weights measure regularity, and weak Fourier series extend these constructions to objects more singular than functions.